Jiten Kumar

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leetcode.com/u/cAj0u6haiT

Summary

Software Developer and AI/ML Engineer with hands-on, full-pipeline experience – from backend systems, REST APIs, and full-stack web apps (Python, Java, JavaScript, Flask, Node.js, React) to deep learning and Generative AI applications (CNN, LSTM, transfer learning, Gemini/OpenAI APIs, RAG). Delivered ML systems exceeding 90% validation accuracy, analytics platforms processing 1M+ records, and multiple end-to-end web and GenAI applications, backed by a strong data structures and problem-solving foundation.

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, SQL
- **Backend Development:** Flask, FastAPI, Node.js, Express.js, REST APIs, Authentication, API Integration
- **Frontend Development:** React.js, Tailwind CSS, HTML, CSS, Vite
- **Machine Learning & Data Science:** Supervised/Unsupervised Learning, CNN, LSTM, Transfer Learning, NLP, Time-Series Forecasting, Model Evaluation (Precision, Recall, F1-Score, AUC-ROC)
- **Generative AI:** Google Gemini API, OpenAI API, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Applications
- **Computer Vision:** OpenCV, Image Classification, TensorFlow Lite
- **Libraries & Frameworks:** TensorFlow, Keras, Scikit-learn, Flask, Streamlit, Pandas, NumPy, Matplotlib, Seaborn
- **Data & Visualization:** EDA, Data Cleaning, Feature Engineering, Statistical Analysis, A/B Testing, Power BI, Excel
- **Databases & Deployment:** MySQL, MongoDB (Mongoose), SQLite, Firebase, REST APIs, Docker, CI/CD Pipelines, MLflow, Model Versioning (DVC), Render
- **Data Structures & Problem Solving:** Arrays, Strings, Recursion, Trees, Graphs, Dynamic Programming – 150+ problems solved on LeetCode
- **Tools & Cloud:** Git, GitHub, Google Cloud, VS Code, Jupyter Notebook, Postman

Experience

Data Science & Analytics Intern **April 2026 – August 2026**
Amdox Technologies *Hyderabad, India*

- Built **NeuralRetail**, a retail analytics platform in Python and Power BI analyzing 1M+ transaction records; automated dashboards for sales trends, customer segmentation, and inventory KPIs, cutting manual reporting effort for stakeholders.
- Built **MediVerse AI**, an ML-powered healthcare recommendation platform, and developed an AI chatbot (Gemini API, Flask, REST APIs) for natural-language symptom intake and conversational triage support.

AI/ML Research Intern **May 2026 – August 2026**
IIIT Ranchi *Hybrid*

- Developing **AgriVision-XAI**, a next-generation plant disease detection system using deep learning, transfer learning, and computer vision techniques.
- Enhancing model accuracy, generalization, and deployment readiness through architecture optimization, experimentation, and research-driven evaluation.

AI/ML Intern **April 2025 – October 2025**
CodexIntern *Hyderabad, India*

- Built the initial version of **Nimbus AI**, a conversational assistant using the Google Gemini API with persistent memory, voice interaction, and automated task workflows.
- Designed a modular Flask backend integrating multiple third-party APIs (weather, web search, crypto) for context-aware, multi-turn conversations.

Full Stack Developer Intern **December 2024 – January 2025**
EliteTech *Remote*

– Developed responsive full-stack web applications using React.js, Express.js, MongoDB, and REST APIs following modern client-server architecture.
– Implemented authentication, CRUD operations, and API integration while collaborating on application testing and deployment.

Machine Learning Engineer (Freelance) **May 2024 – December 2024**
Independent Contract – College-Commissioned Competition Project *Hyderabad, India*

– Built an end-to-end CNN-based crop disease classification system trained on 70K+ labelled images, achieving 90%+ validation accuracy using data augmentation and transfer learning.
– Owned the full ML pipeline – preprocessing, augmentation, model training, and evaluation – using MLflow for experiment tracking and DVC for dataset/model versioning, with a CI/CD pipeline automating deployment to a client-ready inference service.

Projects

Wonderful Crown – Healthcare Recommendation System [Repo] [Live] *Python, Flask, Scikit-learn, SQLAlchemy*

– Built an ML-powered healthcare platform for symptom-based disease prediction with confidence scoring, medicine recommendations, and prediction history tracking.
– Implemented Flask authentication, an admin panel, analytics dashboards, and automated PDF/Excel report exports; designed the database schema with SQLAlchemy.

Cindrix-AI – Generative AI Assistant with Persistent Memory [Repo] [Live] *Python, Flask, Gemini API, RAG, Three.js*

– Built an AI assistant powered by Google Gemini with persistent conversational memory, Retrieval-Augmented Generation (RAG), voice interaction, and real-time weather/cryptocurrency lookup.
– Developed a modular Flask backend for scalable AI applications and an immersive particle-sphere interface using Three.js for an engaging user experience.

StockMentor – AI Stock Market Analysis Platform [Repo] *Python, LSTM, Tiingo API, Matplotlib*

– Built a stock market analysis platform using the Tiingo API for real-time data and an LSTM model for price prediction, achieving **85% prediction accuracy** and driving a **30% increase in user engagement**.
– Added personalized investment tips, watchlists, and alerts to help beginner investors make informed decisions, with 100-day/200-day moving-average feature engineering.

CrowdPulse – AI Crowd Analytics Platform [Repo] *Python, Computer Vision, Deep Learning*

– Built a computer-vision-based crowd analytics platform for real-time crowd detection, density estimation, and occupancy monitoring, targeted at smart-city and public-safety use cases.
– Designed for high-traffic environments such as transportation hubs and large-scale events, surfacing live occupancy analytics for safety monitoring.

BizLink – Buyer & Seller Matching Platform [Repo] *React, Vite, AI Document Analysis*

– Built a B2B platform connecting business buyers and sellers with role-based onboarding, buyer profile browsing, and AI-powered document analysis for compatibility scoring.
– Implemented dark/light theme support and a scalable, responsive design; deployed on Vercel.

Facial Recognition Attendance System [Repo] *Python, OpenCV, Firebase*

– Built a facial recognition system for automated attendance tracking, achieving **95% accuracy** in face detection and recognition.
– Integrated with Firebase for real-time data management, providing a reliable, scalable solution for institutions and organizations.

AgriVision-XAI – Explainable AI for Plant Disease Detection [Repo] *Python, EfficientNetB0, CBAM, Grad-CAM, TensorFlow Lite, Streamlit*

– Built a research-grade explainable AI system for plant disease detection using EfficientNetB0 with CBAM attention, incorporating Grad-CAM, Grad-CAM++, and LIME for model interpretability.
– Added disease severity estimation, cross-domain evaluation, and TensorFlow Lite deployment with an interactive Streamlit app; an explainable, upgraded evolution of an earlier CNN-based Plant Disease Detection model.

Perfume Shop – Full-Stack E-Commerce App [Repo] *React, Tailwind CSS, Vite, Node.js, Express, MongoDB*

- Built a full-stack e-commerce web app with a React/Tailwind/Vite frontend and a Node.js/Express backend, using MongoDB (Mongoose) for product and review data.
- Implemented a product catalog with detail pages, size variants, a review system, and native Web Share API integration.

JobPilot – LinkedIn Application Automation Bot *Python, Selenium*

- Built a Selenium-based automation bot across five core modules to handle LinkedIn Easy Apply flows, including ARIA-based dropdown handling and dynamic form fields.
- Implemented cookie-based session reuse to avoid repeated logins, with configuration support for running across multiple machines.

Other Notable Projects: Voyage-AI – AI travel planner with itinerary/budget generation (Google AI Studio) | Legal AI Assistant – Gemini-powered legal document Q&A (Python) | AI Voice Assistant – 98% speech-recognition accuracy (Python) | Rubik's Cube Solver – Kociemba algorithm with Tkinter GUI | Hand Tracking & Eye Tracking apps (OpenCV, Mediapipe) | Sentiment Analysis App (Python, NLP) | ToDo List App (React.js, Tailwind CSS)

Education

Jawaharlal Nehru Technological University Hyderabad **2021 – 2025**
B.Tech – Computer Science and Engineering (AI & ML Specialization) *CGPA: 7.44*

Kendriya Vidyalaya Gachibowli **2019 – 2021**
Higher Secondary (CBSE) *CGPA: 8.2*

Certifications & Achievements

- **1st Place – Mind Plug Fiesta** *National-Level Hackathon*
- **Python for Data Science** *NPTEL*
- **Build with AI Bootcamp** *Google for Developers*
- **Generative AI Virtual Internship** *Infosys Springboard, 2025*
- **GfG 160 – 160 Days of Problem Solving** *GeeksforGeeks*
- **Gen AI Exchange Program** *Google Cloud Skills Boost*